AI and You

UNDERSTANDING AND USING ARTIFICIAL INTELLIGENCE

SCHH Computer Club Sun City Hilton Head · 2025

Part 1

INTRODUCTION & KEY CONCEPTS

~30 minutes

AI Is Already in Your Life

You've been using AI for years — probably without realizing it

- **Email spam filter** — AI decides what's junk before you ever see it
- **Netflix & Spotify** — AI learns your taste and surfaces what you'll like
- **GPS navigation** — AI reroutes you around traffic in real time
- **Autocorrect** — AI predicts what word you meant to type
- **Credit card fraud alerts** — AI flags unusual charges instantly
- **Voice assistants** — Siri, Alexa, Google — all AI-powered

> The chatbot-style AI getting attention today is a new, more *visible* form of something that has quietly been part of daily life for years.

What Is Artificial Intelligence?

AI is a broad term for computer systems that can do things that normally require human intelligence

- Understanding language
- Recognizing faces, images, and speech
- Making decisions and predictions
- Learning from experience and improving over time

> Think of it less like a robot brain and more like a **very sophisticated pattern-matching machine** trained on enormous amounts of human knowledge.

AI isn't one single thing — it's a *family of technologies*

Generative AI

A type of AI that **creates** new content — rather than just analyzing or sorting existing content

It can generate:

- Text — articles, letters, summaries, code, poetry
- Images — photos, illustrations, artwork
- Audio — music, voice, sound effects
- Video — synthetic footage, animation

> When you ask an AI to write a letter, summarize an article, or create an image of a sunset — that's generative AI at work.

The "generative" part means it **produces something new each time**, not just retrieving a stored answer

Large Language Models (LLMs)

The technology **behind most of today's AI chatbots**

- Trained on vast amounts of text — books, websites, articles, and more
- Learned language by identifying patterns across billions of examples
- Can understand and generate human language naturally
- "Large" refers to the scale of both training data and mathematical complexity

> Think of it like someone who has read an enormous amount becoming a strong writer and communicator — they've absorbed the patterns of language without memorizing every book word-for-word.

ChatGPT, Claude, and Gemini are all built on LLMs

What Is a Chatbot?

An AI you talk to — **in plain English**

- You type a question or request; it responds in full sentences
- No special commands, no search syntax — just conversation
- Very different from older chatbots (phone trees, website pop-ups)
- Today's chatbots understand nuance, context, and follow-up questions

> You talk to it the way you'd talk to a knowledgeable friend.

Quick show of hands — how many of you have tried a chatbot in the past six months?

Context & Memory

WITHIN A SINGLE CONVERSATION

The AI remembers **everything said so far** — you can refer back and it follows along

BETWEEN CONVERSATIONS

Fresh start every time — when you close the chat and open a new one, it remembers nothing

THE ANALOGY

> A brilliant assistant with excellent short-term memory and no long-term memory. Brilliant for the length of a conversation. Blank slate tomorrow.

PRACTICAL TIP

Give the AI context upfront — it doesn't know who you are, your situation, or why you're asking. Instead of *"What should I do about Medicare?"* try *"I'm 72, recently moved to South Carolina, and have Parts A and B. I'm wondering if I need a supplemental plan..."*

Thinking Mode vs. Regular Mode

REGULAR MODE

Fast, conversational — the AI responds immediately, drawing on its training *Great for: everyday questions, writing help, casual conversation*

THINKING MODE

The AI works through the problem **step by step** before answering — like showing its work *Takes a little longer, but handles complex questions more reliably*

> Like the difference between a quick reply and asking someone to sleep on it and give you their considered answer.

Use regular for: simple questions, writing help, casual chat

Use thinking for: planning, analysis, anything with multiple steps or a right answer that matters

Hallucinations

When AI **confidently states something that is simply not true**

- AI generates language based on patterns — not by looking facts up
- It can produce false information that sounds completely authoritative
- A made-up statistic. A fake quote. A book that doesn't exist. A law that was never passed.
- It's not lying — **it doesn't know it's wrong**

> This is the most important thing to understand about AI safety.

THE RULE

Never rely on AI alone for anything important — medical, legal, financial, or factual. Always verify with a trusted second source.

Think of AI as a very well-read research assistant who occasionally makes things up with total confidence.

Knowledge Cutoff

AI models are trained up to a **specific date** — after which they have no knowledge of world events

- If something happened after that date, the AI genuinely doesn't know
- It might guess, or confidently describe something that has since changed
- This is why AI might not know about recent news, new products, or changes in the law

> Always ask about time-sensitive topics with this in mind — and verify with current sources.

SOME AI TOOLS CAN SEARCH THE WEB

Perplexity and others connect AI to live search — giving you current information with sources cited

We'll see this in Part 2

Prompts & Prompting

A **prompt** is simply what you type — your question or request

The way you ask shapes what you get

TIPS FOR BETTER PROMPTS

- **Be specific** — vague questions get vague answers
- **Give context** — tell it who you are and why you're asking
- **Ask for a format** — "explain it simply" / "give me a short list" / "write it as a letter"
- **Follow up** — if the answer isn't right, say so. It responds to feedback.

> Think of it as a conversation, not a search box.

Before: *"Tell me about Medicare."*

After: *"I'm 72 and recently moved to South Carolina. I have Parts A and B but I'm confused about whether I need a supplemental plan. Can you explain my options in plain language?"*

Privacy — What Happens to What You Type?

WHAT YOU SHOULD KNOW

- Most AI tools **store your conversations**, at least temporarily
- Free tools are more likely to use conversations to train future models
- Paid or enterprise versions typically offer stronger privacy protections
- Different companies have different policies — worth reading for tools you use regularly

THE PRACTICAL RULE

Don't share sensitive personal information with any AI tool:

- Social Security numbers
- Financial account numbers or passwords
- Detailed medical records
- Anything you wouldn't say to a helpful stranger

> Treat it like a knowledgeable professional you've just met — useful to talk to, but not someone you'd hand your private documents to.

Deepfakes

AI can generate **realistic fake video, audio, and images** of real people

- A public figure saying something they never said
- A phone call that sounds like a family member in distress
- A photograph of an event that never happened

THE GRANDPARENT SCAM — UPDATED

AI can now **clone a voice** from just a few seconds of audio. Scammers use this to call grandparents pretending to be a grandchild in trouble — needing bail money, emergency funds, urgent help.

> If you receive an unexpected call from a "family member" in crisis — hang up and call them directly on a number you already know.

WATCH FOR

Faces that look slightly off · teeth and hair edges that blur · unnatural blinking

AI Slop

The flood of **low-quality, AI-generated content** now polluting the internet

- Fake news articles written by AI with no human author
- AI-generated product reviews that all sound identical
- Spam blog posts that fill the web with plausible-sounding nonsense
- Fake photos of events, places, and people that don't exist
- Social media posts generated in bulk to manipulate opinion

> You've probably already encountered it without knowing what to call it.

HOW TO SPOT IT

- **Generic, vague writing** that says a lot without saying anything specific
- **No author name** or a publication you've never heard of
- **Reviews that all sound the same** — suspiciously enthusiastic, no specifics
- **Images that look polished but feel slightly wrong** — trust that instinct

THE HABIT THAT PROTECTS YOU FROM BOTH DEEPFAKES AND AI SLOP

Slow down and verify before you believe or share.

Part 2

TOOLS & PRACTICE

~30 minutes · Live demos

Chatbots - The Main Players

Three leading AI chatbots — all capable, each with its own strengths

CHATGPT

OpenAI The name most people know. Widest recognition, broad capabilities, large user community.

CLAUDE

Anthropic Known for thoughtful, nuanced responses. Particularly strong at writing and analysis.

GEMINI

Google Integrated directly into Gmail, Google Docs, and other Google services you already use. If you use Google, you likely already have access.

> Think of them like different brands of a reliable car — Toyota, Honda, Chevrolet. All get you where you're going. Different feel, different fans. Competition keeps them all improving.

All three have free versions. Paid tiers (~$20/month) unlock faster, more capable models.

Getting Better Answers — Prompting Tips

A prompt is simply what you type. The way you ask shapes what you get.

THE BASICS

- **Be specific** — vague questions get vague answers
- **Give context** — tell it who you are and why you're asking
- **Ask for a format** — *"explain simply" / "give me a short list" / "write it as a letter"*
- **Follow up** — say "that's not quite right" and it adjusts

> Think of it as a conversation, not a search box.

Before: *"Tell me about Medicare."*

After: *"I'm 72 and recently moved to South Carolina. I have Medicare Parts A and B but I'm not sure if I need a supplemental plan. Can you explain my options in plain language, focusing on what matters most for someone in good health?"*

The second prompt gets a focused, personalized, useful answer. The first gets a textbook.

Perplexity — AI + Search

The safest starting point for questions about the real world

WHAT MAKES IT DIFFERENT

- Combines AI reasoning with **live web search**
- Cites its sources — you can see exactly where the information came from
- Solves the knowledge cutoff problem — it knows about recent events
- Results are easier to verify than a plain chatbot answer

BEST FOR

- Current events and recent news
- Health and medical questions *(then verify with your doctor)*
- Local information — restaurants, services, businesses
- Any question where you want to know the source

> *"I want to understand something, and I want to know where that understanding comes from."*

ChatGPT & Claude — Conversational AI in Depth

Best for: writing, explaining, brainstorming, analyzing, drafting

WHAT THEY DO WELL

- Write or improve letters, emails, and documents
- Explain complex topics in plain language
- Help plan — trips, events, decisions
- Answer follow-up questions and adjust based on feedback
- Summarize long documents or articles you paste in

LIVE DEMO IDEAS

- Write a letter to an insurance company disputing a claim
- Explain a medical term a doctor used
- Plan a week of healthy dinners based on dietary needs
- Summarize and simplify a complex document

> **Tip:** paste in text and ask it to summarize, simplify, or explain it — one of the most powerful everyday uses

Image Generation — AI as a Visual Tool

AI can now create images from a text description alone

TODAY'S TOOLS

- **ChatGPT** (via GPT-4o) — generates photorealistic and illustrated images from prompts
- **Google Gemini** — similar capability, integrated with Google's ecosystem
- **Adobe Firefly** — designed for practical creative use, commercially safe

WHAT PEOPLE USE IT FOR

- Creating cards, flyers, and invitations
- Visualizing home decor ideas before committing
- Illustrating stories or memories
- Fun — seeing what your prompt produces

YOUR TURN

Suggest a scene and we'll generate it live

> **Important:** AI image tools can also create realistic fake photos of real people — this is the technology behind many deepfakes.

NotebookLM — Talk to Your Own Documents

Google's NotebookLM lets you upload documents and have a conversation with them

WHAT IT DOES

- Upload PDFs, articles, Google Docs, or web pages
- Ask questions about the content in plain English
- Get summaries, explanations, and answers drawn directly from your documents
- **Audio overview** — generates a natural-sounding podcast-style discussion of your material

PRACTICAL USES FOR YOUR AUDIENCE

- Upload Medicare documents and ask "what does this actually mean?"
- Upload HOA rules and ask "what are the rules about fences?"
- Upload a medical report and ask for a plain-English explanation
- Upload research on a topic you care about and explore it conversationally

> The audio podcast feature is particularly remarkable — paste in a document and get a two-voice discussion of it in minutes.

Where to Start

TODAY, TRY ONE OF THESE

1. **Perplexity** *(perplexity.ai)* — search a question you've been wondering about. Check the sources.
2. **Claude or ChatGPT** — ask for help drafting a letter or email you've been putting off.
3. **NotebookLM** *(notebooklm.google.com)* — upload a confusing document and ask it to explain the key points.

REMEMBER

- **Verify anything important** — AI is a starting point, not a final answer
- **Don't share sensitive personal information**
- **Follow up** — if the first response isn't quite right, say so
- **It's okay to experiment** — you can't break it

> *The best way to learn these tools is to use them. Start with a question you actually have.*

Thank You

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